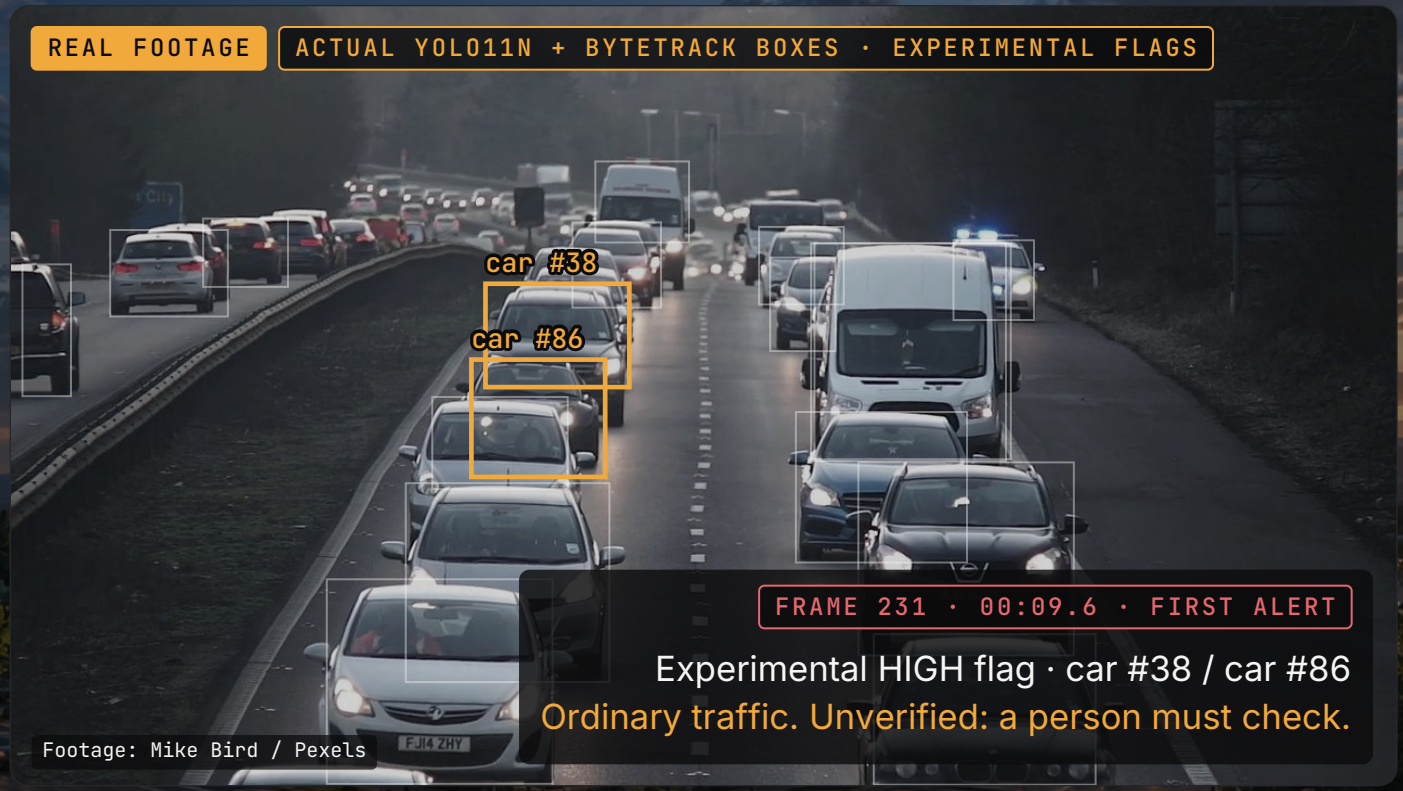


A near miss disappears *in seconds.*

Its evidence shouldn't.

Illustrative image · not analyzed footage

WHAT NEARMISS DOES

Turns fixed-camera traffic footage into *evidence a person can check.*

01
Predict
Projects tracked paths and flags potential conflicts.

02
Detect
Separately screens for contact plus a motion response.

03
Verify
A person reviews the source frame and clip.

04
Notify
Confirmation creates one local, simulated notice.

BUILT BY

Srivibhav Padakandla · Aadi Shukla

WHO IT IS FOR

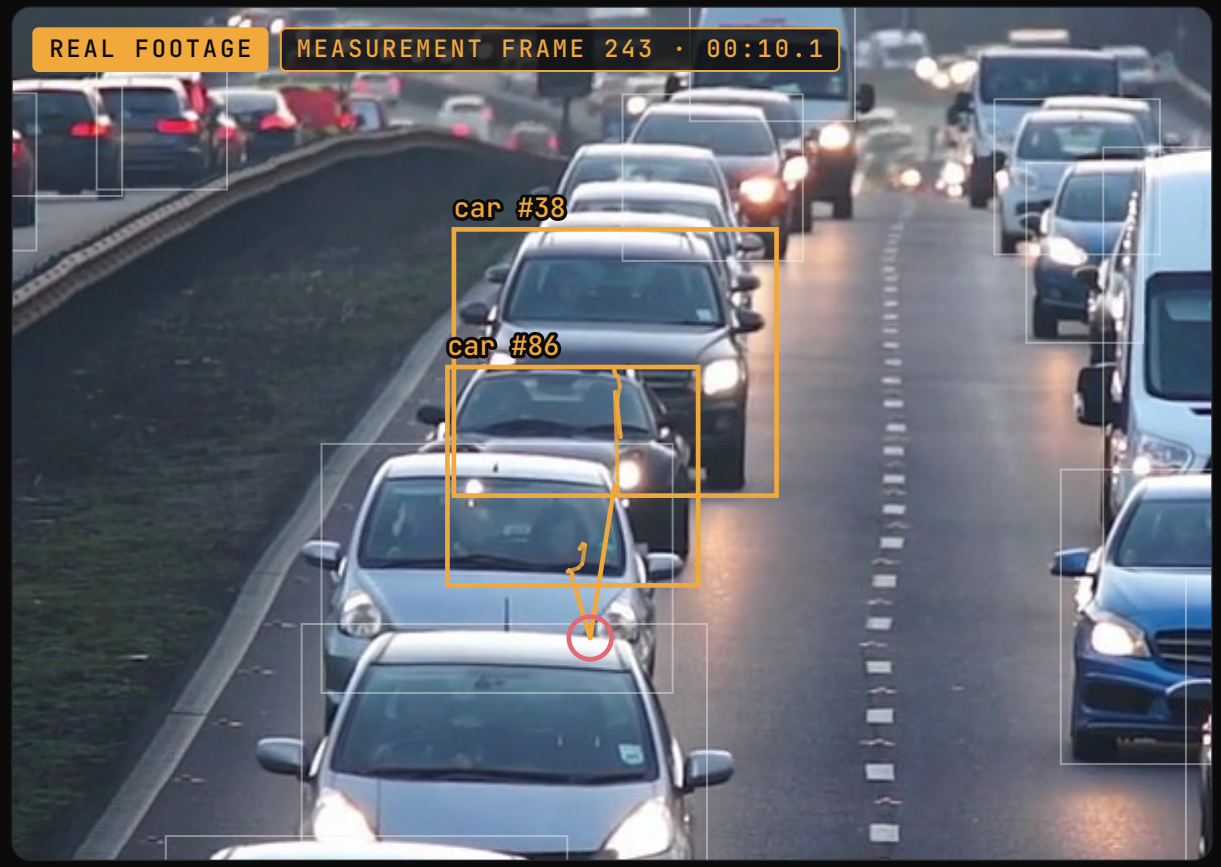
People who review intersections: safety staff, and neighbors preparing a street-safety conversation.

REAL FOOTAGE

Ordinary highway traffic (Mike Bird / Pexels), no incident labels. Boxes and flags are actual NearMiss output: experimental, unverified, human review required.

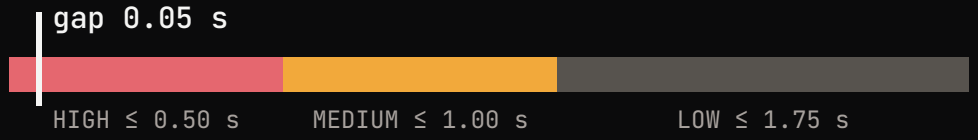
HOW A WARNING IS EARNED

Two projected arrival times. *One gap.*



car #38 reaches the shared point in 1.15 s
car #86 reaches it in 1.20 s
arrival gap 0.05 s → HIGH

Measured on frame 243 (00:10.1). First alert frame 231 (00:09.6). #86 box from frame 242 (latest observation).



HIGH: a trailing vehicle closed on a slower vehicle ahead

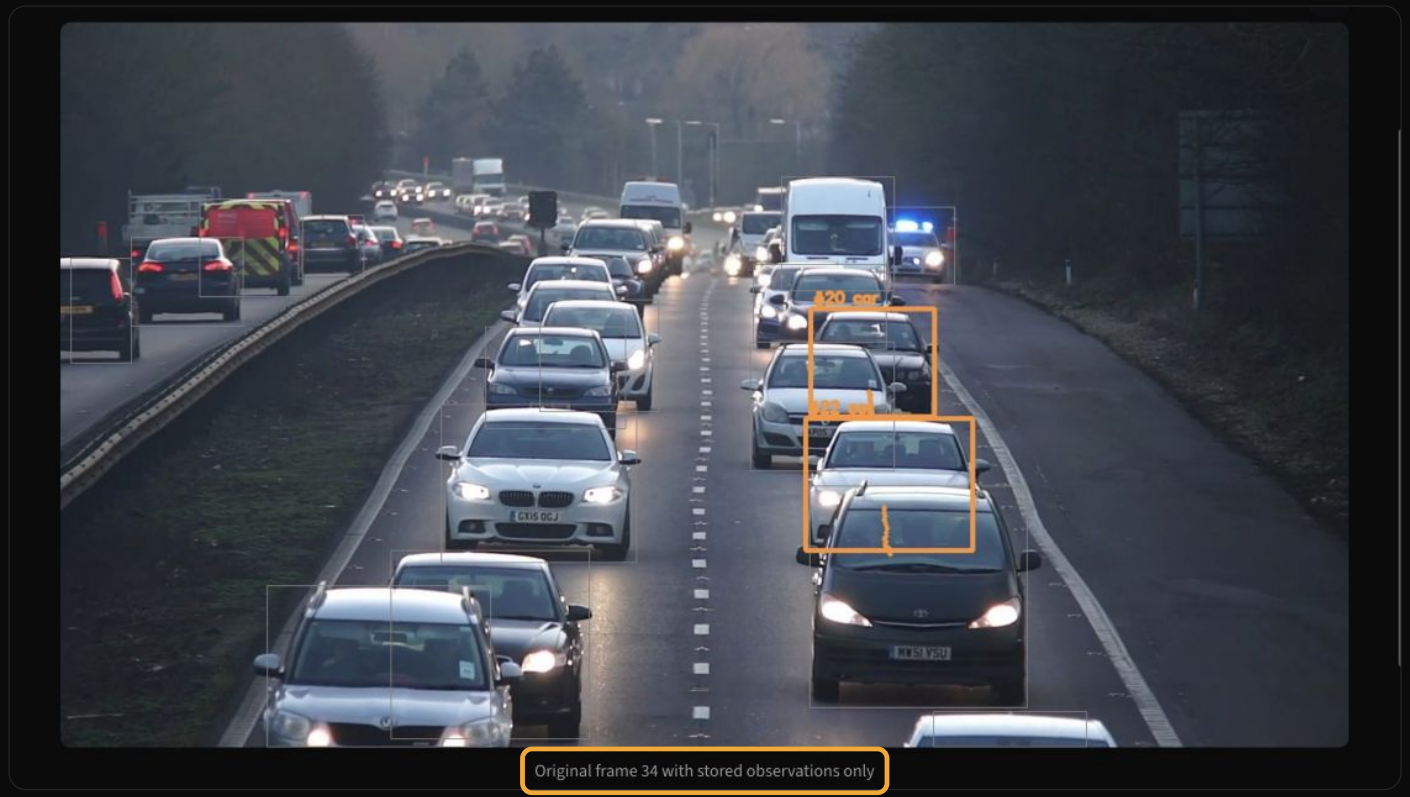
- At constant speed their footprints would meet in about 1.17s, inside the 2.0s look-ahead window.
- Time headway is 0.05s (HIGH ≤ 0.50s, MEDIUM ≤ 1.00s, LOW ≤ 1.75s).

Actual explain_event() output · the app's "Why each warning was raised" text

Image pixels, constant-velocity projection: in perspective, two cars in one lane can "meet". Not a measured speed, distance or probability.

INSPECT THE SOURCE

Every flag points back to *the exact frame.*



Actual app · Investigation · original frame 34, tracks #20 and #22, stored observations only

41 warning candidates 7 HIGH · 16 MEDIUM · 18 LOW
5 suspected contacts 0 confirmed
78 tracker IDs not a vehicle count

00:01.5 HIGH
Opposing / turning paths · car #20 / car #22
Opened in Investigation (frame 34)

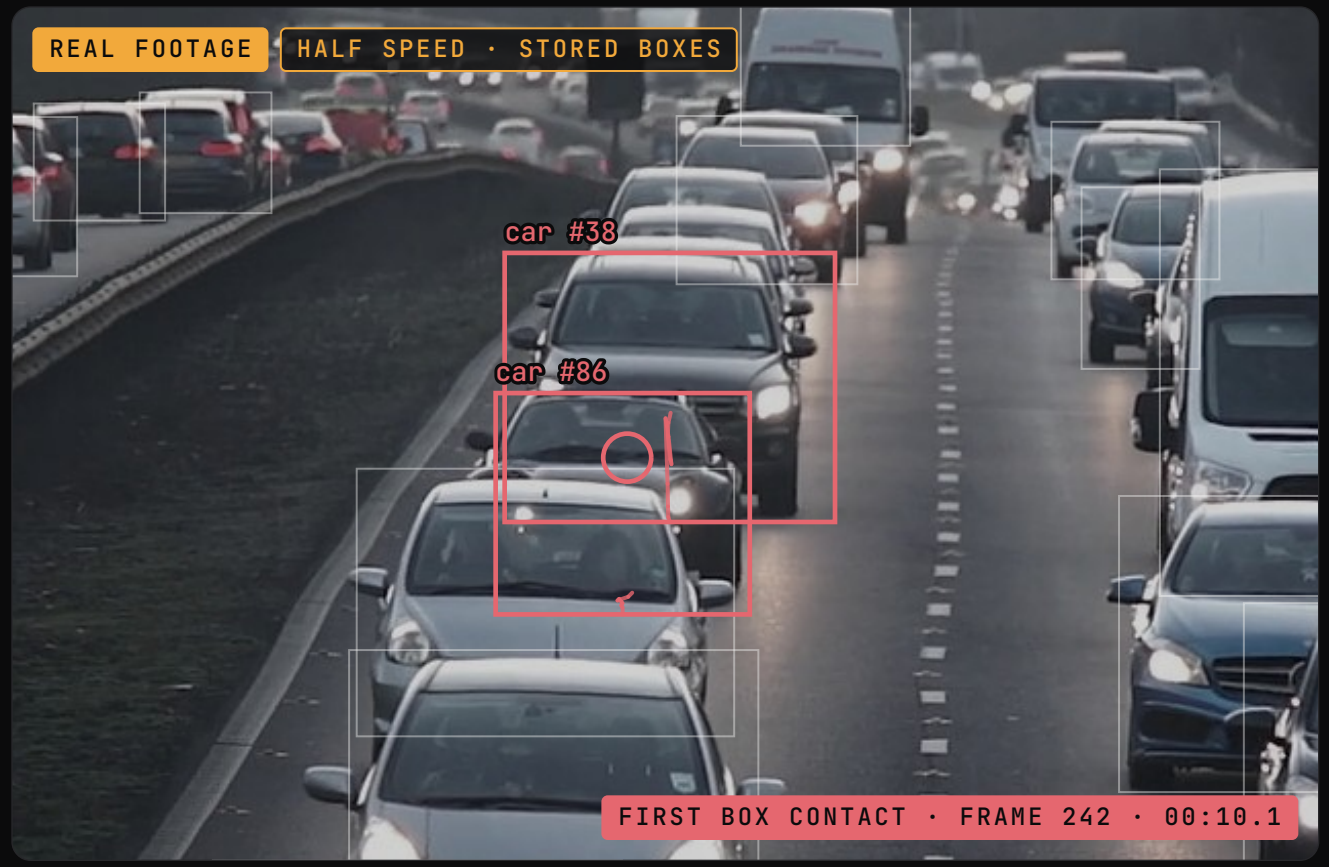
00:09.6 HIGH
Same-direction following · car #38 / car #86
arrival gap 0.05 s · slide 3

00:10.1 CONTACT
Suspected contact · car #38 / car #86
5 signals · unverified · slide 5

Investigation redraws the stored boxes on the original frame; nothing is re-detected. Shown: frame 34 for cars #20 and #22, whose HIGH flag first fired on frame 35.

HUMAN REVIEW

Contact needs more than overlap. *A person decides.*



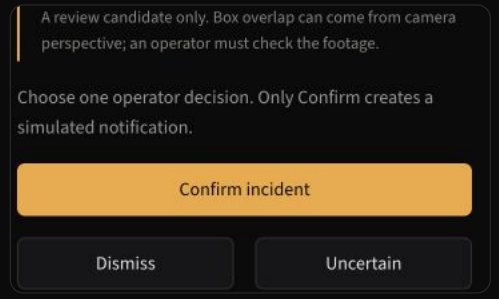
- Persistent box contact
- Significant box overlap
- Converging before contact
- Abrupt motion change
- Sudden deceleration

Warning conflict-029 came 0.46 s before first contact. Correlated measurements, not independent votes.

OUR READING OF THE FRAME · NOT A RECORDED REVIEW

Box “contact” lasted 116 frames (4.8 s). #86 is the nearer car in the same lane, partly hiding #38: overlap from perspective. It looks like queued traffic, not contact.

App: 5 candidates Uncertain · 0 confirmed · 0 notices



Actual app · Incidents review controls on this clip

FROM EVIDENCE TO FOLLOW-UP

No confirmation, no notice. *A plan anyone can check.*

DISPATCH · ONLY AFTER CONFIRM

Candidate → Person confirms → Local notice

Simulated: nothing leaves the computer.

5 candidates 0 confirmed 0 notices



Actual app · Street Lab summary for this clip

STREET LAB · CROSSING ARITHMETIC

Typical walking speed · 3.5 ft/s

19.1 s now

13.4 s hypothetical

Slower walking speed · 2.5 ft/s

26.0 s now

18.0 s hypothetical

Stated assumptions: 60.0 → 40.0 ft crossing · start-up 2.0 s

Arithmetic on entered assumptions. Not a road-safety measurement or crash reduction.

ACTUAL EXPORT · COMMUNITY BRIEF

NearMiss Community Review Brief

> Real traffic footage: “Traffic Flow In The Highway” by Mike Bird. Bundled sample, not your upload.

Evidence summary

- Vehicles tracked: 78; pedestrians tracked: 0
- Potential-conflict warnings: 41 (HIGH 7, MEDIUM 16, LOW 18)
- Suspected incidents: 5 (confirmed 0, dismissed 0, uncertain 5)

Boundaries

- No crash records, city data, or real incidents are used. Counts below come only from the active video's processed detections and this operator's reviews.
- Crossing times are arithmetic on the assumptions entered above (distance ÷ walking speed + start-up time). They illustrate time spent in the roadway; they are not a measured crash reduction or a prediction of safety outcomes.

Downloaded from Street Lab on this clip; no review decisions changed.

WHAT WE BUILT · VERIFIED · HAVE NOT PROVEN

Open-source eyes. *Our own judgment layer. Honest limits.*

BUILT

EXISTING OPEN SOURCE

- Ultralytics YOLO: detects road users
- ByteTrack: keeps IDs across frames
- OpenCV, Streamlit, Plotly, ReportLab

NEARMISS CUSTOM LOGIC

- Trajectory prediction and conflict gates
- Same-direction following check
- Multi-signal suspected-collision detector
- Evidence association and de-duplication
- Review record and simulated dispatch audit trail
- Investigation, briefs, Street Lab, run library

VERIFIED

- All 271 automated tests pass (local, 2026-10-25). They check code behaviour, not detector accuracy
- Replay from the cached tracks reproduces the app's 41 warnings and 5 contact candidates exactly
- Clip, tracks and metadata bound by SHA-256 in the app manifest; the deck checks them
- Real-footage label and credit carried on frames, pages and exports
- Confirm, dismiss and uncertain states covered by tests (SQLite review record)

NOT YET PROVEN

- Accuracy on real footage: not measured. This clip has no ground-truth labels
- All 41 warnings and 5 contact candidates are unverified; perspective and occlusion cause false alarms
- 78 tracker IDs are not a vehicle count
- Image pixels only: no metres, speeds or probabilities
- No crash-reduction claim. Dispatch is simulated

PROPOSED NEXT STEP

A supervised pilot *before any claims.*

01

Permissioned footage from one fixed camera

02

Compare NearMiss flags with human-labelled events to measure accuracy

03

Calibrate the camera view so pixels can become metres

AI ASSISTANCE

OpenAI Codex and Anthropic Claude (Opus) assisted with writing, reviewing and testing code, and with building this deck. Results shown come from running the app and its tests.

FOOTAGE · OPEN SOURCE

"Traffic Flow In The Highway", Mike Bird / Pexels (Pexels License). Ultralytics YOLO · ByteTrack · OpenCV · Streamlit · Plotly · ReportLab · GSAP · Inter, Instrument Serif, JetBrains Mono (OFL)

A near miss disappears *in seconds. Its evidence shouldn't.*

NearMiss · Srivibhav Padakandla · Aadi Shukla

Illustrative image · not analyzed footage